10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Review Exercise 10 — Practical Geometry of Circles

Unit 10 | Review + MCQs | New Book (PECTAA)

Review Exercise 10 (Solved)

The review covers locating the centre of a circle, completing a circle from an arc, and every case of drawing a tangent.

1. Four possible answers are given for the following questions. Choose the correct answer:

(i) In the adjacent figure, ABC is:

(a) a chord (b) an arc
(c) a tangent (d) a secant

Answer: (c) a tangent — the line touches the circle at B and cuts it nowhere else

(ii) There are _____ types of arcs.

(a) 2 (b) 3
(c) 4 (d) 5

Answer: (a) 2 — minor arc and major arc (a semi-circle is the boundary case)

(iii) Right bisector of the chord of a circle always passes through the:

(a) diameter (b) non-collinear points
(c) radius (d) centre

Answer: (d) centre — the perpendicular from the centre bisects every chord

(iv) A circle has only one:

(a) chord (b) centre
(c) diameter (d) secant

Answer: (b) centre — a circle has one fixed centre, but many chords and diameters

(v) The point where two tangents meet outside a circle forms:

(a) a semicircle (b) a diameter
(c) a radius (d) an angle

Answer: (d) an angle — the two tangent segments enclose an angle at that point

(vi) Two equal tangents from a point to a circle can be drawn when the point is _____ the circle.

(a) on (b) inside
(c) outside (d) at centre of

Answer: (c) outside — from an outside point the two tangent lengths are equal

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(vii) Tangents drawn from a single external point to a circle are:

(a) unequal in length (b) perpendicular to each other
(c) equal in length (d) inside the circle

Answer: (c) equal in length — tangents from an external point are equal

(viii) At least how many chords are needed to locate the centre of the circle?

(a) 1 (b) 2
(c) 3 (d) 4

Answer: (b) 2 — two right bisectors of two chords meet at the centre

(ix) To draw a tangent at middle point of an arc, first step is to:

(a) join endpoints of an arc (b) draw radius
(c) draw a line perpendicular to chord (d) draw an angle

Answer: (a) join endpoints of an arc — the chord is needed first, then its right bisector gives the radius at the mid-point

(x) The angle between the radius and a tangent at the point of contact is:

(a) 30° (b) 60°
(c) 90° (d) 120°

Answer: (c) 90° — the radius is perpendicular to the tangent at the point of contact

2. Draw a circle of radius 1.4 cm and draw a tangent at point P, when P lies on its circumference.

1. Take O as centre and radius 1.4 cm, and draw the circle.
2. Mark P on the circumference and join O to P.
3. At P draw a perpendicular to OP with the help of compasses.
4. Produce the perpendicular on both sides of P; it is the required tangent.

Answer: Circle of radius 1.4 cm drawn with the tangent at P

3. Draw a circle of radius 1.2 cm and draw a tangent at point P, when P is at a distance of 5 cm from the centre.

1. Take O as centre and radius 1.2 cm, and draw the circle.
2. Produce a radius outwards and mark P on it such that OP = 5 cm.
3. Find the midpoint M of OP.
4. Take M as centre and radius mOM = mMP and draw a semi-circle; it cuts the circle at T.
5. Join P to T and produce it; PT is the required tangent.

Answer: Tangent drawn from P, 5 cm away from the centre

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4. Draw a circle of radius 1.7 cm. Draw two tangents that meet an angle of 30°.

1. Draw a circle of radius 1.7 cm with centre O and draw a diameter AOB.
2. At A draw a perpendicular to OA; this is the first tangent.
3. At O draw m∠AOC = 30°, taking C on the circle.
4. At C draw a perpendicular to OC; it meets the first tangent at P.
5. PA and PC are the required tangents, meeting at m∠APC = 30°.

Answer: Two tangents drawn enclosing an angle of 30°

5. Take a part of the circular track, use chords and perpendicular bisectors to complete a circular track.

1. Take three points A, B and C on the given part of the track.
2. Join AB and BC.
3. Draw the right bisector of AB.
4. Draw the right bisector of BC.
5. The two right bisectors meet at O, the centre of the track.
6. Draw the circle with centre O and radius OA = OB = OC; the track is completed.

Answer: The circular track is completed with centre O

6. Two decorative fences touch the circular flower bed of radius 2.1 cm and meet outside it at an angle of 30° to form an entrance arch. Draw two tangents to the flower bed from the point where the fences meet at an angle of 30°.

1. Draw the circular flower bed with centre O and radius 2.1 cm, and draw a diameter AOB.
2. At A draw a perpendicular to OA; this is the first fence.
3. At O draw m∠AOC = 30°, taking C on the circle.
4. At C draw a perpendicular to OC; it cuts the first fence at P.
5. PA and PC are the two required tangents, meeting at m∠APC = 30°.

Answer: Two tangents drawn meeting at 30°, forming the entrance arch

Quick Revision — Unit 10 at a Glance

Centre of a circle = meeting point of the right bisectors of any two chords.

One and only one circle passes through three non-collinear points.

An arc can be completed with equal chords and equal angles, without the centre.

Tangent at a point on the circle: perpendicular to the radius there.

Tangent from an outside point: semi-circle on OP gives the point of contact.

Two tangents enclosing θ: diameter, tangent at one end, angle θ at the centre.